
Informational Actualization Model

A Note on Entropic Gravity and the Thermodynamic-Gravity Conjecture

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Abstract

The Informational Actualization Model (IAM) is a cosmological extension of the Jacobson–Cai–Kim thermodynamic gravity framework. Rather than modifying the entropy functional S_{BH} on the cosmic horizon — the approach taken in Barrow and Tsallis holographic cosmologies — IAM identifies a physically motivated additional entropy source, S_{info} , arising from gravitational decoherence of collapsing matter. This entropy enters the Clausius relation and propagates to the perturbation equations as a single Hubble friction term with no new free parameters: the coupling constant $\beta_m = \Omega_m/2$ is derived from the virial theorem, and the activation function $E(a) = \exp(1 - 1/a)$ from the decoherence integral over cosmic history. This note situates IAM within the entropy-gravity framework familiar from holographic dark energy research and identifies the key structural difference between *entropy-modification* approaches and *entropy-source* approaches.

1. The Shared Starting Point

IAM begins from the same foundation as holographic dark energy models in the Barrow–Tsallis tradition: the *thermodynamic-gravity conjecture*. In this framework the standard Friedmann equations governing cosmic expansion are not fundamental — they are thermodynamic

equations of state derivable from the first law of thermodynamics applied to the cosmic horizon.

The derivation proceeds through the Cai–Kim (2005) formalism. The apparent horizon of a Friedmann–Robertson–Walker universe has area $A_H = 4\pi/H^2$, Gibbons–Hawking temperature $T_H = H/2\pi$, and Bekenstein–Hawking entropy $S_{\text{BH}} = A_H/4$. Applying the first law

$$dE = T_H dS_{\text{BH}} - W dV \quad (1)$$

recovers the standard Friedmann equations exactly. Gravitational field equations are thermodynamic identities [Jacobson, 1995].

This starting point is shared by IAM, Barrow holographic dark energy, Tsallis holographic dark energy, and Kaniadakis cosmology. All of these frameworks accept the thermodynamic-gravity conjecture. They diverge in what they do next.

2. Two Roads From the Conjecture

Once the thermodynamic-gravity conjecture is accepted there are two natural ways to extend it beyond Λ CDM:

2.1. Road 1 — Modify the entropy functional

Barrow entropy replaces S_{BH} with a fractal-deformed expression $S_{\text{Barrow}} = (A/A_0)^{1+\Delta/2}$, introducing a deformation parameter Δ that quantifies quantum-gravitational fractal structure on the horizon surface [Saridakis, 2020a]. Tsallis entropy replaces S_{BH} with a non-extensive expression $S_{\text{Tsallis}} = \gamma A^\delta$, where δ captures long-range correlations [Saridakis et al., 2018]. In both cases the modification is to the geometric entropy term itself, and the new parameter (Δ or δ) must be constrained by data.

These are well-motivated approaches. They explore what happens when the entropy-area law receives corrections — a natural question if the horizon geometry is not smooth at the Planck scale.

2.2. Road 2 — Identify an additional entropy source

IAM takes a different path. The geometric entropy S_{BH} remains standard Bekenstein–Hawking. Instead, IAM identifies a physically motivated additional contribution to the total horizon

entropy budget:

$$S_{\text{total}} = S_{\text{BH}} + S_{\text{info}}, \quad (2)$$

where S_{info} arises from *gravitational decoherence* of collapsing matter.

The reasoning is as follows. Collapsing matter continuously generates classical information: quantum states decohere into classical configurations as gravitational gradients cause wave-function branching. Each decoherence event is irreversible and carries an entropy cost of $k_B \ln 2$ per bit (Landauer’s principle, experimentally verified by Bérut et al. 2012 and Jun et al. 2014). This information is holographically encoded on the cosmic horizon at the Gibbons–Hawking temperature $T_H = H/2\pi$.

S_{info} is not a modification to the entropy-area law. It is a *source term* — entropy entering the horizon accounting from a physical process that standard thermodynamic gravity had not previously included.

3. Where the Two Roads Lead

The distinction between Road 1 and Road 2 is not merely taxonomic — it has direct observational consequences.

In Barrow and Tsallis cosmologies the entropy modification alters the Friedmann equations — the background expansion history is modified. The new parameter (Δ or δ) affects both the expansion rate $H(z)$ and the growth of structure simultaneously, and must be fitted to data. Recent constraints from DESI DR2 BAO data confirm that both Barrow and Tsallis cosmologies are compatible with observations but find that neither resolves the H_0 tension, and that both are disfavoured relative to Λ CDM by information criteria due to the additional free parameter [Luciano et al., 2025].

In IAM, S_{info} is sourced by matter decoherence and therefore enters the perturbation equations only — specifically as an additional Hubble friction term in the matter growth equation:

$$\delta'' + \left[2 + \beta_m E(a)\right] H \delta' - \frac{3}{2} \Omega_m H^2 \delta = 0, \quad (3)$$

where primes denote derivatives with respect to cosmic time. The background expansion remains standard Λ CDM. The photon sector is unaffected.

This is the origin of the dual-sector μ – Σ prediction. Massive particles travel on timelike paths; proper time elapses; gravitational decoherence acts continuously. Photons travel on

null geodesics; no proper time elapses; they are never subject to the decoherence process. The prediction $\Sigma = 1$ (no lensing-potential modification) follows from this geometric fact, not from a parameter choice. The prediction $\mu < 1$ (suppressed matter growth) follows from the friction term in Eq. (3).

4. The Coupling Constant $\beta_m = \Omega_m/2$

The coefficient β_m in Eq. (3) is not a free parameter. It is derived from the virial theorem. For any self-gravitating system bound by a $1/r$ potential:

$$2\langle T \rangle + \langle V \rangle = 0 \implies \langle T \rangle = \frac{1}{2}|\langle V \rangle|. \quad (4)$$

Exactly half the gravitational energy budget resides in kinetic degrees of freedom. In IAM the kinetic channel is the decoherence-producing channel — irreversible random motions of particles within collapsing structures generate the classical information requiring holographic encoding. The informational energy density contribution is therefore $\Omega_m/2$, giving $\beta_m = 0.1575$.

Six independent N-body simulation groups have measured the mass-weighted virial ratio $2K/|U|$ directly [Bryan & Norman, 1998, Bett et al., 2007, Neto et al., 2007, Ludlow et al., 2010, Power et al., 2012, Klypin et al., 2016], with combined result $\langle \eta_{\text{virial}} \rangle = 0.815 \pm 0.025$, giving $\beta_m = 0.159 \pm 0.010$. This agrees with the virial-theorem-derived $\Omega_m/2 = 0.1575$ to 1.1%. The same 1/2 partition holds from hydrogen atoms through stellar structure through galaxy clusters — 37 orders of magnitude in physical scale.

The Planck 2018 MCMC posterior independently recovers $\beta_m = \Omega_m/2$ to within 0.2σ without this value being fitted. That independent confirmation from the most constraining cosmological dataset is the single most compelling quantitative result in the framework.

5. The Activation Function $E(a)$

$E(a)$ is derived by integrating the decoherence rate over cosmic history, weighted by the Gibbons–Hawking encoding temperature $T_H = H/2\pi$ and horizon area $A_H = 4\pi/H^2$. The result is:

$$E(a) = \exp\left(1 - \frac{1}{a}\right). \quad (5)$$

Two boundary conditions fix $E(a)$ completely without free parameters:

- $E(1) = 1$: the present-day informational contribution is normalized to the matter energy density scale.
- $E(a) \rightarrow e$ as $a \rightarrow \infty$: the self-limiting nature of the feedback (expansion dilutes matter, slows decoherence) causes cumulative entropy to asymptote to a finite ceiling.

At $a = 1$ today, $E(1)/e = 1/e \approx 36.8\%$, placing the current epoch at the *inflection point* of the activation function where the rate of informational entropy production is at its peak. Six published structure-formation analyses find an effective growth exponent $n_{\text{eff}} = 3.22 \pm 0.44$, consistent with the range 3.0–3.5 required by the decoherence integral.

6. Comparison With Barrow/Tsallis Holographic Cosmologies

Table 1 summarizes the structural comparison. Both families of models emerge from the same thermodynamic-gravity conjecture. The key difference is where new physics enters: in the entropy functional (Barrow/Tsallis) versus in the entropy source term (IAM).

Table 1: Structural comparison of entropy-based cosmological frameworks.

Feature	Barrow / Tsallis HDE	IAM
Foundation	Thermodynamic-gravity conjecture (Jacobson; Cai–Kim)	Same foundation
Modification	Entropy functional: $S_{\text{BH}} \rightarrow S_{\text{Barrow}}$ or S_{Tsallis}	Entropy source: $S_{\text{total}} = S_{\text{BH}} + S_{\text{info}}$
Free parameters	Δ (Barrow) or δ (Tsallis) — fitted to data	None — $\beta_m = \Omega_m/2$ from virial theorem
Background modified?	Yes — Friedmann equations altered	No — standard Λ CDM background
Perturbations modified?	Yes (through modified Friedmann equations)	Yes — direct friction term only
μ – Σ signature	Generally $\mu \neq 1$, $\Sigma \neq 1$	$\mu < 1$, $\Sigma = 1$ exactly
H_0 tension	Not resolved [Luciano et al., 2025]	Predicted: 72.5 km/s/Mpc (output)
AIC vs. Λ CDM	Disfavoured (extra parameter)	Indifferent (no extra parameter)

The AIC comparison deserves particular attention. Because Barrow and Tsallis cosmologies introduce one additional free parameter, information criteria penalise them relative to Λ CDM even when the fit improves. IAM introduces no additional free parameter — β_m is fully determined by Ω_m , which is already a standard Λ CDM parameter. IAM therefore carries no AIC penalty.

7. A Note on the Entropy Exponent

There is a natural question to ask of the Barrow/Tsallis framework: why does the entropy exponent take the value it does? The deformation parameters Δ and δ are physically motivated — fractal horizon structure, non-extensive statistics — but their values are not derived from first principles. They are measured from data.

IAM suggests a perspective on an adjacent question. If S_{info} from decoherence is the correct physical source of the cosmological entropy correction, then the apparent need for a non-standard entropy functional in other models may reflect an incomplete accounting of entropy *sources* rather than a modification to the entropy-area law itself. Standard Bekenstein–Hawking entropy may be exact; the horizon entropy budget simply includes more terms than previously recognised.

Whether Barrow or Tsallis entropy captures genuine Planck-scale physics is an open question IAM does not resolve. The narrower point is that the same late-time cosmological phenomenology that motivates entropy-modification approaches can be obtained from an entropy-source approach with zero free parameters and a derivable coupling constant.

8. MCMC Validation and Key Predictions

8.1. Validation

IAM has been validated against the Planck 2018 CMB likelihood through 17 converged MCMC chains (12 Level 1 via MGCAMB μ – Σ parametrisation; 3 Level 2 via direct Fortran modification of CAMB’s perturbation equations; 2 exploratory background runs confirming the perturbation-level mechanism). All chains satisfy $R - 1 < 0.01$ Gelman–Rubin convergence.

Table 2: Key IAM predictions and current observational status.

Quantity	IAM Value	Current Data	Source
μ_0	-0.136 (derived)	0.05 ± 0.22	DESI 2025
Σ_0 (dev.)	0 (null worldlines)	0.008 ± 0.045	ACT+WMAP+SDSS
σ_8	0.800 (derived)	0.802 ± 0.020	Stölzner et al. 2025
H_0^{matter}	72.5 km/s/Mpc (derived)	73.04 ± 1.04	Riess et al. 2022
$\Delta\chi^2$	+0.54	confirmed	17 Planck chains
β_m	$\Omega_m/2 = 0.1575$	0.159 ± 0.010	6 N-body studies

8.2. Falsifiable predictions

IAM makes three near-term quantitative predictions:

1. **Euclid μ - Σ measurement.** The IAM prediction $\mu_0 = -0.136$, $\Sigma_0 = 0$ is directly testable by Euclid at projected $\sigma(\mu_0) \sim 0.04$ sensitivity. The combination $\mu < 1$ with $\Sigma = 1$ is a unique signature in the μ - Σ plane: $f(R)$ gravity gives $\mu > 1$, $\Sigma > 1$; DGP braneworld gives $\mu > 1$; general Horndeski theories predict $\Sigma \neq 1$.
2. **DESI DR2 growth rate trajectory.** The IAM $f\sigma_8(z)$ trajectory, suppressed by the $\beta_m E(a)$ friction term, is consistent with DESI DR2 growth rate data without requiring $w < -1$. A direct confrontation of the DESI DR2 $f\sigma_8$ measurements against the IAM predicted trajectory tests the interpretation that the apparent phantom crossing at $z \sim 0.5$ is a growth-geometry tension, not a dark energy signature.
3. **Galaxy cluster lensing-to-dynamics ratio.** The dual-sector structure predicts $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu_{\text{eff}}(z) > 1$ in a redshift-dependent pattern that approaches unity at high z as $\mu(a) \rightarrow 1$. This is qualitatively distinct from a constant hydrostatic bias and testable with current eROSITA, Planck SZ, and weak lensing datasets in combination.

If Euclid measures μ consistent with zero at $\sigma(\mu_0) \sim 0.04$ precision, IAM is ruled out. No other currently published modified gravity model predicts $\mu < 1$ with $\Sigma = 1$ from first principles.

9. Summary

The shared foundation is the Jacobson–Cai–Kim result: gravitational field equations are thermodynamic equations of state derivable from the first law applied to the cosmic horizon.

IAM’s specific contribution is to identify gravitational decoherence of collapsing matter as a physical *source* of additional horizon entropy. This source term, derived via Landauer’s principle and the virial theorem, produces a matter-sector Hubble friction term with no free parameters. The coupling $\beta_m = \Omega_m/2$ is recovered independently by six N-body simulation groups and by the Planck 2018 MCMC posterior to 0.2σ .

The framework is consistent with the full Planck 2018 likelihood ($\Delta\chi^2 = +0.54$), predicts $\sigma_8 = 0.800$ in agreement with current weak lensing measurements, and places the H_0 tension endpoints within 1σ of both Planck and SH0ES with no parameter adjustment. It carries no AIC penalty because no new free parameter is introduced.

Euclid’s μ – Σ measurement at projected $\sigma(\mu_0) \sim 0.04$ sensitivity provides a direct test of the unique IAM prediction $\mu_0 = -0.136$, $\Sigma_0 = 0$. The full derivation chain, MCMC chains, modified CAMB Fortran source, and analysis scripts are publicly available at the Zenodo DOI and GitHub repository listed on the title page.

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